



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to SRI POONGAVANATHAMMAN AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

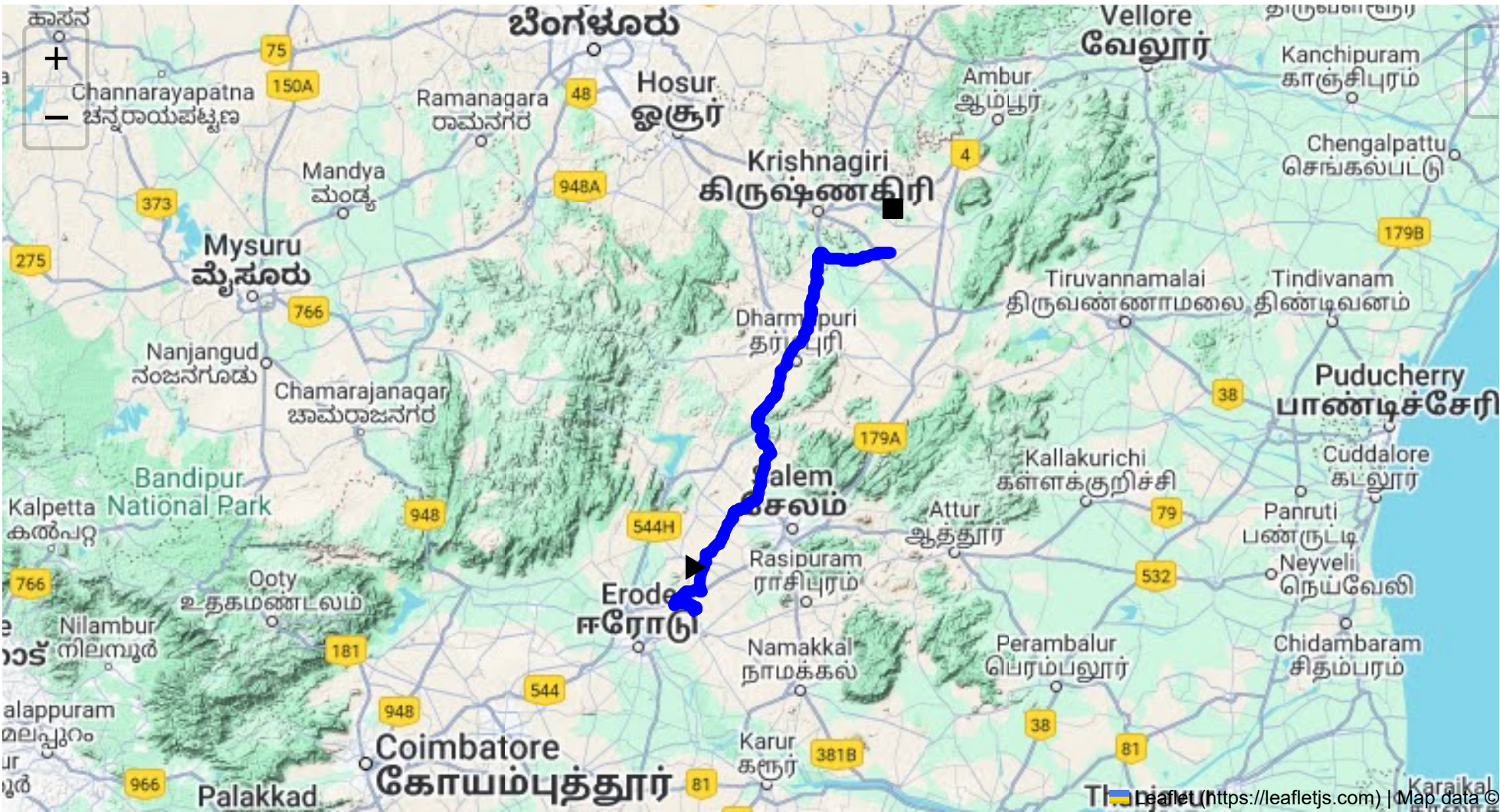
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 164.57 km
Estimated Duration: 3.3 hours
Adjusted Duration (Heavy Vehicle): 4.2 hours
Start: (11.4381, 77.8734)
End: (12.417073, 78.422561)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The journey spans approximately 164.57 kilometers starting at Sangagiri, Tamil Nadu, and concluding in Sevvampatti, Tamil Nadu. Key waypoints include:

- Sankari RS - Erode Privu Rd

- Bhavani Main Road (at two points)
- Salem
- SH 86 areas (Erumaipatti, Konganapuram, Koranampatti)

The route predominantly traverses through rural areas with occasional urban intersections.

2. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu generally experiences a tropical climate:

- **Summer (March to June):** High temperatures reaching up to 40°C can make vehicle operation challenging, affecting driver endurance and increasing risks of tire blowouts.
- **Monsoon (July to September):** Heavy rains can cause road flooding and reduced visibility, especially in rural stretches without proper drainage.
- **Winter (October to February):** Relatively calmer weather, but foggy conditions in the mornings can impact visibility.

Weather-related hazards are primarily associated with the monsoon season.

3. Traffic Patterns, Highlighting Peak Hours and Congestion-Prone Areas

- **Morning (7 AM - 9 AM) and Evening Rush Hours (5 PM - 7 PM):** Moderate congestion near urban stretches, particularly around Salem.
- **Market Areas:** The Bhavani Main Road segments near local trading hubs might face congestion intermittently.
- **Weekend Traffic:** Higher probability of increased traffic, especially near tourist attractions around Salem.

4. Assessment of Road Quality and Infrastructure

- **Good Quality Roads:** Highways such as SH 86 are generally well-paved and maintained.
- **Rural Roads:** Can be narrow with limited shoulders and occasional potholes, requiring careful navigation.

5. Suggestions for Alternative Routes for Emergencies

- In case of severe delays or roadblocks, detouring through Salem provides several alternate highways and better access to facilities.
- Utilizing the NH544 bypass around Erode can help avoid urban congestion.

6. Local Regulations Affecting Hazardous Material Transport

- **Restricted Hours:** Some urban areas and highways may have restrictions on heavy vehicle movement during daylight hours.
- **Permits:** Ensure all transport permits specific to hazardous materials are secured in advance.

7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials

- **Typical Incidents:** Common incidents involve overturned vehicles due to speeding or improper load securing.
- **Local Accident Zones:** Bhavani Main Road has documented a few incidents, mainly due to driver error or poor road conditions during monsoons.

8. Environmental Considerations and Sensitive Areas

- **Protected Areas:** Be cautious near ecological zones, particularly nearby reserves which could have movement restrictions.
- **Pollution Control:** Adhere to all pollution standards as per regional environmental guidelines.

9. Analysis of Communication Coverage

- **Good Coverage:** Near urban centers like Salem.
- **Potential Dead Zones:** Some rural segments, particularly through less-populated stretches, might experience weak mobile network signals.

10. Estimated Emergency Response Times for Different Route Segments

- **Urban Proximity:** Near Salem, response times can be 15-30 minutes.
- **Rural Areas:** Could range from 45 minutes to over an hour due to distance and road conditions.

12. Overall Summary of Risk Assessment

This route involves moderate risk, with potential hazards primarily from weather conditions, road quality, and occasional traffic congestion. Ensuring proper preparation with respect to weather forecasts, obtaining necessary permits, and familiarizing with alternative routes can mitigate many of these risks. Constant communication and awareness of emergency protocols are fundamental for drivers dealing with hazardous materials.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43968, 77.87345	15 KM/Hr
1	Turn	High	11.44029, 77.87544	15 KM/Hr
2	Turn	High	11.44900, 77.87406	15 KM/Hr
3	Turn	High	11.45352, 77.85698	15 KM/Hr
4	Turn	Medium	11.45764, 77.85219	30 KM/Hr
5	Turn	High	11.46303, 77.84945	15 KM/Hr
6	Turn	High	11.45571, 77.81702	15 KM/Hr
7	Turn	Medium	11.45654, 77.81706	30 KM/Hr
8	Turn	Medium	11.46916, 77.82431	30 KM/Hr
9	Blind Spot	Blind Spot	11.49225, 77.89560	10 KM/Hr
10	Turn	High	11.49453, 77.88488	15 KM/Hr
11	Turn	Medium	11.56348, 77.90077	30 KM/Hr
12	Turn	Medium	11.57547, 77.90567	30 KM/Hr
13	Turn	Medium	11.59189, 77.90645	30 KM/Hr
14	Turn	Medium	11.59323, 77.90796	30 KM/Hr
15	Turn	Medium	11.63857, 77.94831	30 KM/Hr
16	Turn	High	11.64207, 77.95199	15 KM/Hr
17	Turn	Medium	11.64708, 77.95119	30 KM/Hr
18	Turn	Medium	11.67896, 77.96882	30 KM/Hr
19	Turn	Medium	11.69776, 77.97983	30 KM/Hr
20	Turn	High	11.70233, 77.97869	15 KM/Hr
21	Turn	Medium	11.71394, 77.99572	30 KM/Hr
22	Turn	Medium	11.71916, 78.01117	30 KM/Hr
23	Turn	Medium	11.72504, 78.01680	30 KM/Hr
24	Turn	Medium	11.73766, 78.03445	30 KM/Hr
25	Turn	High	11.73651, 78.03868	15 KM/Hr
26	Turn	Medium	11.74120, 78.04588	30 KM/Hr
27	Turn	Medium	11.78005, 78.05788	30 KM/Hr
28	Turn	Medium	11.86928, 78.08384	30 KM/Hr
29	Turn	Medium	11.88009, 78.08164	30 KM/Hr
30	Turn	Medium	11.91277, 78.05853	30 KM/Hr
31	Turn	Medium	11.93375, 78.06378	30 KM/Hr
32	Turn	Medium	11.93751, 78.05325	30 KM/Hr
33	Turn	Medium	11.95551, 78.04706	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
34	Turn	High	11.95865, 78.04991	15 KM/Hr
35	Turn	High	11.96021, 78.04945	15 KM/Hr
36	Turn	Medium	11.96281, 78.05124	30 KM/Hr
37	Turn	High	11.96442, 78.05104	15 KM/Hr
38	Turn	High	11.96557, 78.04930	15 KM/Hr
39	Turn	High	11.96657, 78.04930	15 KM/Hr
40	Turn	Medium	11.96752, 78.05031	30 KM/Hr
41	Turn	Medium	12.17054, 78.16831	30 KM/Hr
42	Turn	Medium	12.22535, 78.19168	30 KM/Hr
43	Turn	Medium	12.24696, 78.19569	30 KM/Hr
44	Turn	Medium	12.25143, 78.20019	30 KM/Hr
45	Turn	Medium	12.25445, 78.20013	30 KM/Hr
46	Turn	Medium	12.29202, 78.19967	30 KM/Hr
47	Turn	Medium	12.32029, 78.20737	30 KM/Hr
48	Blind Spot	Blind Spot	12.40701, 78.21763	10 KM/Hr
49	Turn	Medium	12.40629, 78.22028	30 KM/Hr
50	Turn	Medium	12.40691, 78.22148	30 KM/Hr
51	Turn	High	12.40669, 78.22256	15 KM/Hr
52	Blind Spot	Blind Spot	12.41749, 78.22507	10 KM/Hr
53	Turn	High	12.39685, 78.30841	15 KM/Hr
54	Turn	Medium	12.39490, 78.31233	30 KM/Hr
55	Turn	High	12.41562, 78.40739	15 KM/Hr
56	Turn	Medium	12.41675, 78.40879	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Sri Shanthi Hospital	11.569272, 77.901937	30 km/h	Medium
2	hospital	Sugumar Hospital	11.7396031, 78.0426955	30 km/h	Medium
3	hospital	Omalur Government Hospital	11.74161, 78.044296	30 km/h	Medium
4	clinic	R.R. Nursing Home	11.7394129, 78.0439861	30 km/h	Medium
5	hospital	Sri Hari Hospital	11.7396726, 78.0440701	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
6	hospital	Sri Sugam Hospital	11.7429821, 78.0461039	30 km/h	Medium
7	hospital	Saravana Hospital	11.7409303, 78.0458308	30 km/h	Medium
8	hospital	Arun Hospital	11.7429892, 78.0466881	30 km/h	Medium
9	clinic	Saratha Clinic	11.7422818, 78.0458175	30 km/h	Medium
11	clinic	Sri Meenakshi Nursing Home, Salem	11.7449162, 78.0474637	30 km/h	Medium
12	hospital	Vinayaka Child Hospital	11.745812, 78.0480225	30 km/h	Medium
13	hospital	Dr. G.R.Thilakam Hospital	11.746097, 78.04769	30 km/h	Medium
14	clinic	Divyam Child Care Clinic	11.7447135, 78.0473858	30 km/h	Medium
15	hospital	Thulasi Hospital	11.7461516, 78.0469984	30 km/h	Medium
16	clinic	Momota Clinic	11.744982, 78.047397	30 km/h	Medium
17	hospital	Amirthalingam Hospital	11.7456708, 78.0470548	30 km/h	Medium
18	hospital	Children Hospital	11.7458377, 78.0480205	30 km/h	Medium
19	clinic	Gomathi Clinic	11.7445293, 78.0470401	30 km/h	Medium
20	clinic	Shri Ramana Poly Clinic	11.7477122, 78.0479293	30 km/h	Medium
21	hospital	Dr. Nagaraj Hospital	11.7464035, 78.0472401	30 km/h	Medium
22	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
24	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
25	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
26	hospital	Thoppur,Government Hospital	11.9398824, 78.0551658	30 km/h	Medium
28	hospital	Government Hospital	12.3048002, 78.2060881	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
10	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
23	college	Dhirajlal Gandhi College of Technology	11.7968014, 78.0562095	30 km/h	Medium
27	college	Laxminarayana Polytechnic College	12.0421277, 78.103203	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
29	school	Government Boys Higher secondary school	12.3047059, 78.2065495	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44900, 77.87406



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45352, 77.85698



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45764, 77.85219



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46303, 77.84945



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45571, 77.81702



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45654, 77.81706



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46916, 77.82431



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.49225, 77.89560



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49453, 77.88488



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.56348, 77.90077



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.57547, 77.90567



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.59189, 77.90645



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.59323, 77.90796



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.63857, 77.94831



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.64207, 77.95199



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64708, 77.95119



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67896, 77.96882



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.69776, 77.97983



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.70233, 77.97869



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71394, 77.99572



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71916, 78.01117



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.72504, 78.01680



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.73766, 78.03445



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.73651, 78.03868



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.74120, 78.04588



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.78005, 78.05788



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.86928, 78.08384



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.88009, 78.08164



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.91277, 78.05853



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93375, 78.06378



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93751, 78.05325



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95551, 78.04706



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.95865, 78.04991



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96021, 78.04945



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96281, 78.05124



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96442, 78.05104



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96557, 78.04930



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96657, 78.04930



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96752, 78.05031



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.17054, 78.16831



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.22535, 78.19168



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.24696, 78.19569



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.25143, 78.20019



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.29202, 78.19967



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.32029, 78.20737



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 12.40701, 78.21763



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.40629, 78.22028



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.40691, 78.22148



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.40669, 78.22256



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 12.41749, 78.22507



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.41562, 78.40739



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.41675, 78.40879